NEURAL INFORMATION PROCESSING SYSTEMS | KTH VETENSKAP OCH KONST | ELECTRONIC ARTS | KBLab

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Why and Where Do Generative Models Fail?

RA-ClipScore: Making Generative Model Evaluation More Interpretable

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Beyond a single score.
A deeper understanding of generative models.

1 Motivation

Global scores hide critical differences.

Conventional metrics summarize generative model performance but cannot explain which attributes differ or where spatial biases occur.



CLIPScore

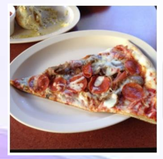
Zebra standing in a grassy landscape.

0.78



RA-CLIPScore

zebra	0.86
grass	0.62
sky	0.56



CLIPScore

A slice of pizza on a plate on table.

0.83

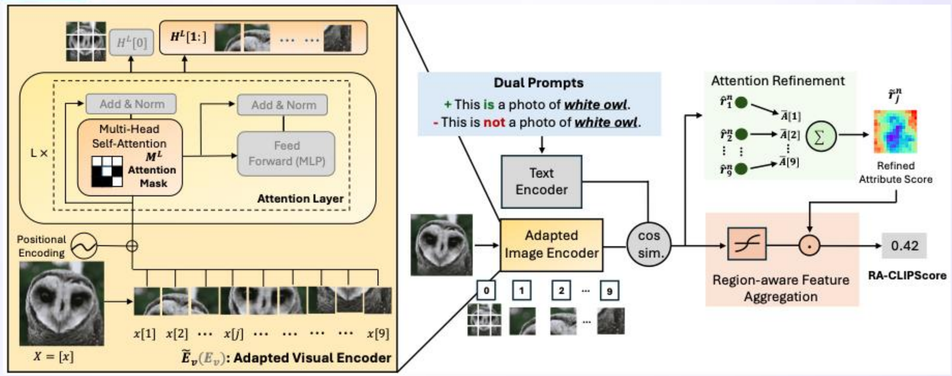


RA-CLIPScore

pizza	0.91
plate	0.81
table	0.64

2 RA-ClipScore

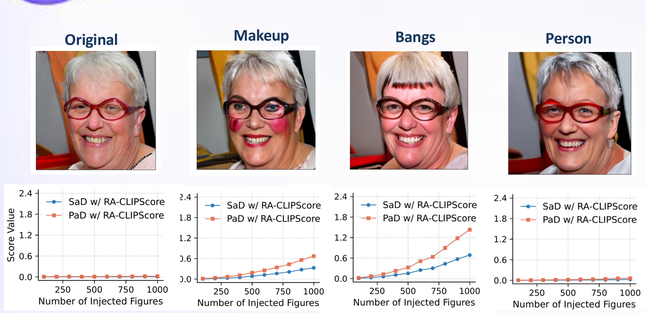
Attribute-aware and spatially grounded evaluation.



- **Attribute-level evaluation** Evaluate each attribute independently.
- **Region-aware representation** Capture fine-grained spatial information.
- **Interpretable comparison** Reveal where and why models differ.

3 Experiments

Comprehensive evaluation results on multiple generative models and datasets.



Injection Experiment. We inject attribute-biased images (makeup and bangs) into FFHQ at increasing ratios. SaD and PaD consistently increase with biased injections while remaining stable under unbiased controls, demonstrating sensitivity to attribute distribution shifts.

Metric	Style GAN2	Style GAN3	LDM (50)	LDM (200)
FID ↓	**2.94**	5.20	8.85	9.53
KID (×10⁻⁴) ↓	**5.0**	8.2	60	69
Precision ↑	**0.792**	0.769	0.703	0.700
Recall ↑	0.665	**0.700**	0.666	0.663
Coverage ↑	**0.948**	0.947	0.832	0.827
Density ↑	**1.11**	1.02	0.72	0.67
SaD (×10⁻³) ↓	**2.25**	2.98	5.64	6.44
PaD (×10⁻³) ↓	**4.9**	6.8	12.3	14.0

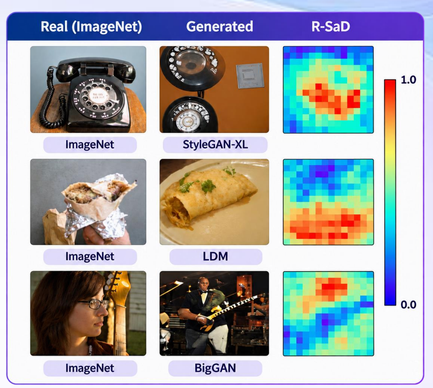
Table 4. Quantitative evaluation results of image generative models on FFHQ.

Setting	Dual Prompts	Local Feat.	Attn. Refine.	Global Token Src.
w/o DP	✗	✓	✓	Layer L
w/o LT	✓	✗	✗	Layer L
w/o AT	✓	✓	✗	Layer L
w/o GE	✓	✓	✓	Layer $L-1$
Ours (Full)	✓	✓	✓	**Layer L**

Table 2. Ablation study configurations. Each variant removes one component at a time from dual prompts, local tokens, attention refinement or global embeddings.

Attribute	CLIPScore	HCS	Ours (Full)	w/o DP	w/o LT	w/o GE	w/o AR
Global (11)	0.83 ± 0.67	0.72 ± 0.43	**1.13 ± 0.91**	0.76 ± 0.49	1.11 ± 0.58	0.95 ± 0.57	0.85 ± 0.70
Facial (3)	1.47 ± 0.65	1.76 ± 0.54	**2.03 ± 0.63**	1.73 ± 0.48	1.62 ± 0.62	1.57 ± 0.63	1.59 ± 0.54
Other (4)	0.93 ± 0.54	0.62 ± 0.53	**0.92 ± 0.76**	0.70 ± 0.58	0.69 ± 0.70	0.75 ± 0.60	0.75 ± 0.64

Table 3. Ablation study on 2D CelebA attributes, grouped into global, facial, and other attribute sets. Results are reported as mean ± standard deviation of Cohen's *d* across attributes. Higher values indicate better separation between positive and negative samples. Best results are highlighted in bold.



Spatial Evaluation on ImageNet. Each row shows a real example (left), a generated sample (middle), and the corresponding R-SaD heatmap (right). Heatmaps visualize regional divergence normalized from 0 (blue, higher occurrence in generated images) to 1 (red, higher occurrence in training data).

4 Key Findings

Our analysis reveals interpretable bias and patterns.

01 Attribute bias
Models perform unevenly across different attributes, with larger gaps in fine-grained details.

02 Spatial inconsistency
Patterns are often spatially localized, revealing systematic region-level biases.

03 Model comparison
Different models excel in different attributes, and our metric provides a unified and interpretable comparison.

From a single score to interpretable evaluation.

Attribute-aware | Region-aware | Explainable | Model Comparison

Project Website →

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